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LETTER OF TRANSMITTAL

House of Representatives
Committee on Science and
Astronautics
Washington, D. C.
24 March, 1961

Honorable Overton Brooks
Chairman, Committee on Science and Astronautics

Dear Mr. Chairman:
I am forwarding herewith for committee consideration a report on "Proposed Studies on the Implications of Peaceful Space Activities for Human Affairs." This report was prepared for NASA under a contract with The Brookings Institution; it has been reviewed by members of the committee staff.

The report is an important effort of the executive branch to comply with section 102(c) of the National Aeronautics and Space Act of 1958 in regard to identification of the long-range goals of the national space program and the gaging of their effect on American society. It is felt

therefore, that the study and dissemination of the report for committee purposes will be useful. Of course, there is no implication here that the report necessarily reflects the views of any member of the committee or staff.

Sincerely Yours,

Charles F. Ducander,
Executive Director and Chief Counsel.

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SUMMARY — ATTITUDES AND VALUES — THE GENERAL PUBLIC — PARA. 4:

Though intelligent or semi-intelligent life conceivably exists elsewhere in our solar system, if intelligent extra-terrestrial life is discovered in the next 20 years, it will very probably be by radio telescope from other solar systems. Evidences of its existence might also be found in artifacts left on the moon or other planets. The consequences for attitudes and values are unpredictable, but would vary profoundly in different cultures and between groups within complex societies; a crucial factor would be the nature of the communication between us and the other beings. Whether or not earth would be inspired to an all-out space effort by such a discovery is moot: societies sure of their own place in the universe have disintegrated when confronted by a superior society, and others have survived even though changed. Clearly, the better we can come to understand the factors involved in responding to such crises the better prepared we will be.

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Contact with Extra-terrestrials

This is from the Congressional hearing report to gage the effects of the long range space goals on American society. The following are the exerpts from this 272 page report which deal with the matter of contact with extra-terrestrials.

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PROPOSED STUDIES ON THE IMPLICATIONS OF PEACEFUL SPACE ACTIVITIES FOR HUMAN AFFAIRS

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by the Brookings Institution

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REPORT OF THE
COMMITTEE ON SCIENCE AND ASTRONAUTICS
U.S. HOUSE OF REPRESENTATIVES
EIGHTY-SEVENTH CONGRESS
FIRST SESSION

GENERAL PUBLIC -- THE IMPLICATIONS OF A DISCOVERY OF EXTRATERRESTRIAL LIFE:

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Recent publicity given to efforts to detect extra-terrestrial messages via radio telescope has popularized--and legitimized--speculations about the impact of such a discovery on human values.[33] It is conceivable that there is semi-intelligent life in some part of our solar system or highly intelligent life which is not technologically oriented, and many cosmologists and astronomers think it very likely that there is intelligent life in many other solar systems. While face-to-face meetings with it will not occur within the next tewnty years (unless its technology is more advanced than ours, qualifying it to visit earth), artifacts left at some point in time by these life forms might possibly be discovered by our space activities on the moon, Mars, or Venus. If there is any contact to be made within the next twenty years, it would be most likely by radio-- which would indicate that these beings had at least our own technological level.

An individual's reactions to such a radio contact would in part depend on his cultural, religious, and social background, as well as on the actions of those he considered authorities and leaders, and their behavior, in turn would in part depend on their cultural, social, and religious environment.[34] The discovery would certainly be front-page news everywhere; the degree of political or social repercussion would probably depend on leadership's interpretation of (1) its own role, (2) threats to that role, and (3) national and personal opportunities to take advantage of the disruption or reinforcement of the attitudes and values of others. Since leadership itself might have great need to gage the direction and intensity of public attitudes, to strengthen its own morale and for decisionmaking purposes, it would be most advantageous to have more to go on than personal opinions about the opinions of the general public and other leadership groups.

The knowledge that life existed in other parts of the universe might lead to a greater unity of men on Earth, based on the "oneness" of man or in the age-old assumption that any stranger is threatening. Much would depend on what, if anything, was communicated between man and the other beings: since after the discovery there will be years of silence (because even the closest stars are several light-years away, an exchange of radio communication would take twice the number of light-years separating our Sun from theirs), the fact that such beings existed might simply become one of the facts of life but probably not one calling for action.[35] Whether earthmen would be inspired to all-out space efforts by such a discovery is a moot question. Anthropological files contain many examples of such societies, sure of their place in the universe, which have disintegrated when they had to associate with previously unfamiliar societies espousing different ideas and different life ways; others that survived such an experience usually did so by paying the price in changes in values and attitudes and behavior.

Since intelligent life may be discovered at any time via the radio telescope research presently underway, and since the consequences of such a discovery are presently unpredictable because of our limited knowledge of behavior under even an approximation of such dramatic circumstances, two research areas can be reccommended--

> Continuing studies to determine emotional and intellectual understanding, and attitudes--and successive alterations of them if any--regarding the possibility and consequences of discovering intelligent extra-terrestrial life.[36]

> Historical and empirical studies of the behavior of peoples and their leaders when confronted with dramatic and unfamiliar events or social pressures.[37] Such studies might help to provide programs for meeting and adjusting to the implications of such a discovery. Questions one might wish to answer by such studies might include: How might such information, under what circumstances, be presented to or withheld from the public for what ends? What might be the role of the discovering scientists or other decisionmakers regarding the release of the fact of discovery?

33 See, for example, the many speculations on Project Ozma. One such: "Project Ozma Begins Operation at National Radio Astronomy Observatory," National Science Foundation Press Release NSF-60-120, 12 April, 1960.

34 The positions of the major American religious denominations, the Christian sects, and the eastern religions on the matter of extr-terrestrial life need elucidation. Consider the following:

"The Fundamentalist (and antiscience) sects are growing apace around the world, and as missionary enterprises, may have schools and a good deal of literature attached to them. One of the important things is that, where they are active, they appeal to the illiterate and semiilliterate (including, as missions, the preachers as well as the congregation) and can pile up a very influential following in terms of numbers. For them, the discovery of other life--rather than any other space product--would be electrifying. Since the main ones among these sects are broadly international in their scope, and are in some places, a news source, the principal distributors of mass media materials, an important source of value interpretation, a central social institution, an educational institution, and so on, some scattered studies need to be made both in their home centers and churches and their missions, in relation to attitudes about space activities and extra-terrestrial life. Additionally, because of international effects of space activities and, in the event of its happening, of the discovery of extra-terrestrial life, even though space activities are not internationalized, it is very important to take into account other major religions. So, for example, Buddhist priests are heavily politically engaged in Ceylon. So, too, in Burma, many politically active men (including U Nu) are professedly active Buddhists. The Burmese covoked the sixth great Buddhist Council which brought together a huge international group of Buddhist lay and ecclesiastical leaders and it seems likely that--at least in the case of Theravada Buddhism--with the wide participation of modern educated, politically active men, Buddhist beliefs and principals are being reinterpreted. We need, and we do not have, good observations and interpretive statements about the possible repercussions of space activities, etc., thr these Buddhists." (Correspondence with Dr. Rhoda Matraux. The observations are based on fieldwork with Montserrat Anthropological Expedition, 1953-54; fieldwork in Haiti; an examination of sectarian literature.)

If plant life or some subhuman intelligence were found on Mars or Venus, for example, there is on the face of it no good reason to suppose these discoveries, after the original novelty had been exploited to the fullest and worn off, would result in substantial changes in perspective or philosophy in large parts of the American public, at least any more than, let us say, did the discovery of the coelacanth, or the panda. It might be well that this sort of discovery would simply not be sufficiently salient for most people most of the time to cause any noticeable shift in philosophy or perspective. If superintelligence is discovered, the results become quite unpredictable. It is possible if the intelligence of these creatures were sufficiently superior to ours, they would choose to have little if any contact with us. On the face of it, there is no reason to believe that we might learn a great deal from them, especially if their physiology and psychology were substantially different from ours.

It has been speculated that, of all groups, scientists and engineers might be the most devastated by the discovery of relatively superior creatures, since these professions are the most clearly associated with the mastery of nature, rather than with the understanding and expression of man. Advanced understanding of nature might vitiate all our theories at the very least, if not also require a culture and perhaps a brain inaccessable to earth scientists. Nature belongs to all creatures, but man's aspirations, motives, history, attitudes, etc., are presumably the proper study of man. It would also depend, of course, on how their intelligence was expressed; it does not necessarily follow that they would excel technologically.

It is perhaps interesting to note that when asked what the consequences of the discovery of superior life would be, an audience of the Saturday Review readership chose, for the most part, not to answer the question at all, in spite of their detailed answers to many other speculative questions. Perhaps the idea is so foreign that even this readership was bemused by it. But one can speculate, too, that the idea of intellectnally superior creatures may be anxiety-provoking. Nor is it clear what would be the reactions to creatures of approximately equal and communicable intelligence to ours.

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What may present a particularly knotty philosophical problem, and one which would seem most clearly to have the potentials of profound repercussions for our values and attitudes and philosophies, could arise if we found a creature whose intelligence and behavior, by our standards, was indeterminate to the point that we were unable to decide whether or not it should be treated morally and ethically as if it were a "human being". Certainly this could provide a continuing subject of controversy across and within various earth cultures; some people who had not otherwise speculated on these matters might gain a sense of complexity of the universe. For a convincing presentation of this idea, see Vercours, "You Shall Know Them", Pocket Books (1955).

On this general problem see Daniel C. Raible, "Rational Life in Outer Space?" America, Vol. 103 (Aug 13, 1960), pp. 532-535; Wolfgang D. Miller, "Religion in Space", Man Among the Stars, Criterion Books, (1957), pp. 221-240; and "Oxnam Sees Space Conquest in 175 Years", Washington Star, 4 January 1960.

35 Professor Jiri Nehnevajsa and Albert S. Francis of Columbia University, in April-June 1960, surveyed samples out of about 100 legislators and 100 university students in both Brazil and Finland. These respondents were asked to indicate which of a series of circumstances they foresaw as changed by a series of developments, including some in space. (In what follows the fugures are given in the following order: (1) Brazilian legislators and (2) students; (3) Finnish legislators and (4) students.) The discovery of civilized alien life was foreseen as increasing the changes for East-West reconciliation by 15, 11, 5, and 6 percent of the respondents, and as increasing the chances of a third political force by 6, 6, 11, and 20 percent of the respondents. Increased status quo and increased liklihood of disarmament accounted for most of the remaining scattered responses, being 2, 5, 21, and 21 percent, and 5, 7, 6, and 5 percent. The remaining types of situations facing the world were seen as essentially unaffected by this event.

36 A possible but not completely satisfactory means for making the possibility "real" for many people would be to confront them with the present speculations about the I.Q. of the porpoise and to encourage them to expand on the implications of this situation. Unfortunately the semantics of "animal", at least for Americans, is such that even a human level I.Q. would not be as threatening as a "being" which wasn't an earth animal.

37 Such studies would include historical reactions to hoaxes, psychic manifestations, unidentified flying objects, etc. Hadley Cantrel's study "Invasion from Mars" (Princeton University Press, 1940), would provide a useful if limited guide in this area. Fruitful understanding might be gained from a comparative study of factors affecting the responses of primitive societies to exposure to technologically advanced societies. Some thrived, some endured, and some died.